Zhengyang Yu

Software Engineering Student

San Diego & Bay Area
+1 510 754 7176
zizha.work@outlook.com
resume.zizhaolaisuimuqu.top
zizhaolaisuimuqu
zhengyang-yu-592a19309
More →

Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA
Expected Graduation: June 2028; GPA: 4.0/4.0
Relevant Coursework: Systems Programming, Computer Organization, Advanced Data Structures, Design and Analysis of Algorithms

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*
- Designed and built standalone C++/FreeRTOS firmware for an ESP32-S3 system integrating an IMU, touch display, network services, and multi-account authentication.
- Implemented concurrent network tasks, adaptive HTTPS polling, device-code authorization, persistent configuration, and bounded structured diagnostics for reliable long-running operation.
- Processed IMU input for orientation-aware rendering and water simulation; debugged timing, networking, state, and display failures on physical hardware using repeatable PlatformIO build and deployment workflows.

2025 **HTML to ePub Converter**, *Personal Project*
- Engineered a Python data-processing pipeline that parses HTML and XML, validates document structure, embeds metadata, and produces standards-compliant ePub files.
- Improved reliability through modular transformations, structured error handling, and repeatable validation of generated output.

2024 **Home Assistant Presence Plugin**, *Personal Project*
- Built a Python integration that models room occupancy from entry and exit sensor events and exposes real-time state for home automation.
- Designed an event-driven REST API workflow with explicit state transitions to support testing, troubleshooting, and reliable smart-lighting behavior.

Skills

Programming C++, Python, Java, C
Embedded ESP32-S3, FreeRTOS, concurrent tasks, networking, IMU and touch integration
Reliability Software testing, hardware validation, structured logging, runtime diagnostics, configuration management
Engineering Object-oriented design, data structures, algorithms, debugging, API design, technical documentation
AI Development Codex-assisted codebase exploration, implementation, refactoring, debugging, and verification
Tools Git, Linux, PlatformIO, Docker, VS Code

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch
Supported community members with computer troubleshooting and taught basic software skills.